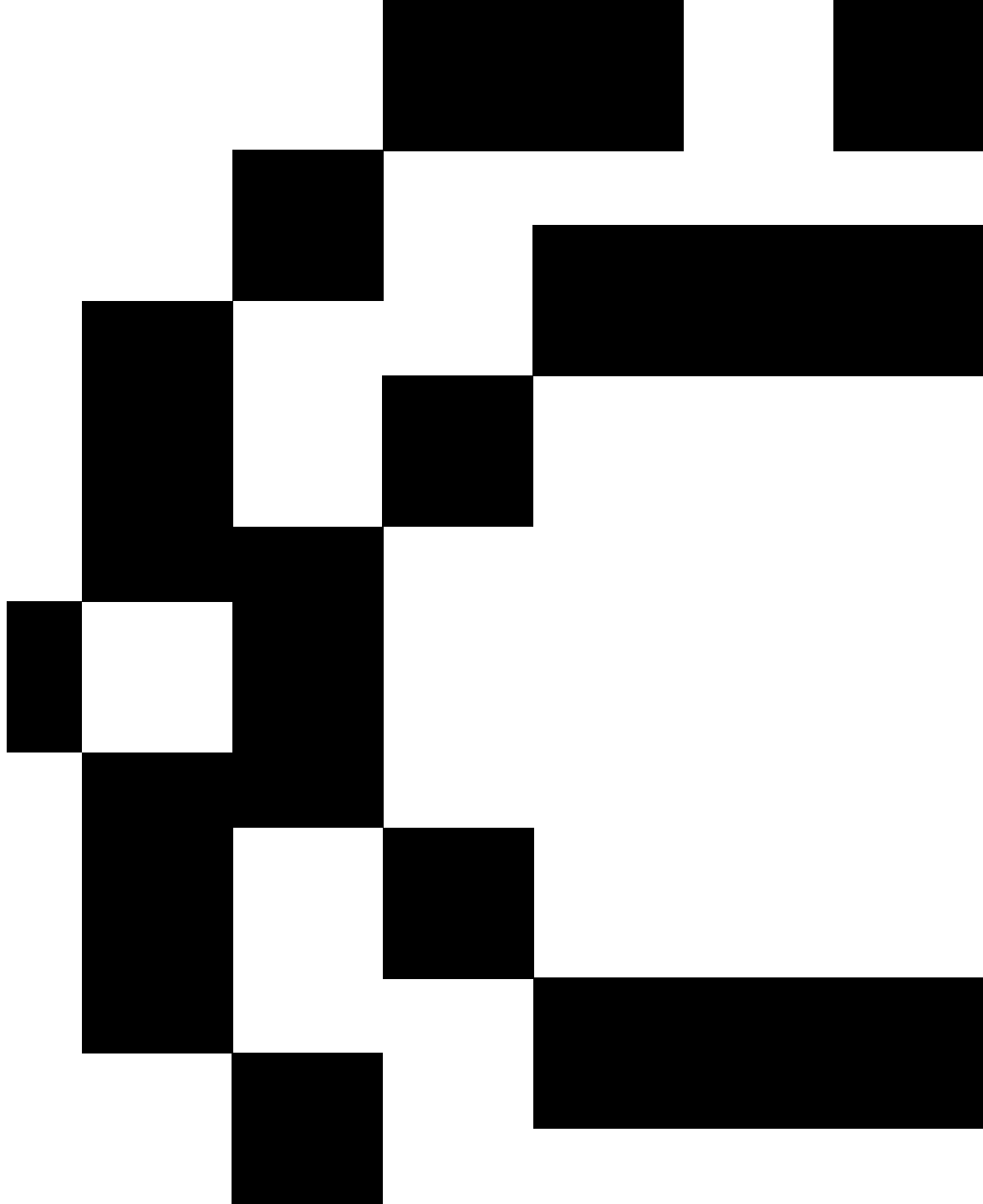
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 03 - String Operations and Type Conversion

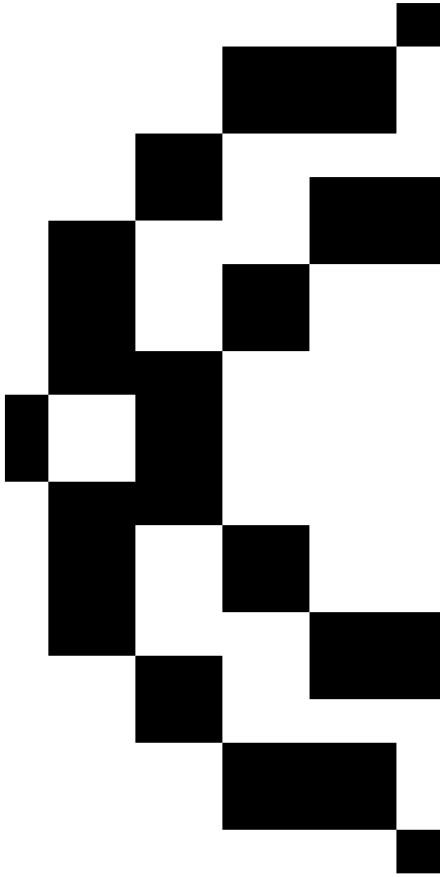


Exercise 1

Define a variable `my_string = "I am learning about strings"`. Extract the word `"learning"` from it and print it. Do this twice: once using positive indexes and once using negative indexes.

Example

- Input: `my_string = "I am learning about strings"`
- Output: `learning`
- `learning`



Exercise 1 — Solution

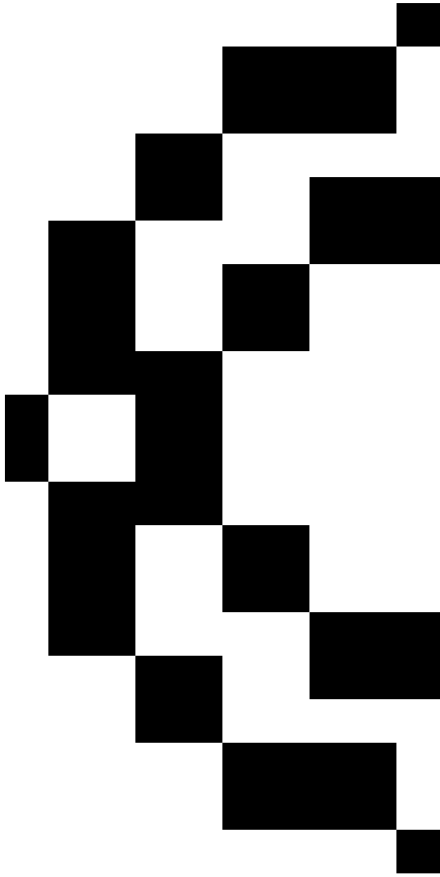
```
my_string = "I am learning about strings"
print(my_string[5:13])    # "learning" — positive indexes
print(my_string[-22:-14]) # "learning" — negative indexes (27-22=5, 27-14=13)
```

Exercise 2

Store the string "`Universidade NOVA de Lisboa`" in a variable `msg`. Print it as-is. Then store the all-uppercase version in a variable `upper_msg` and print it. Finally, store the all-lowercase version in `lower_msg` and print it.

Example

```
Input:  msg = "Universidade NOVA de Lisboa"
Output: Universidade NOVA de Lisboa
        UNIVERSIDADE NOVA DE LISBOA
        universidade nova de lisboa
```



Exercise 2 — Solution

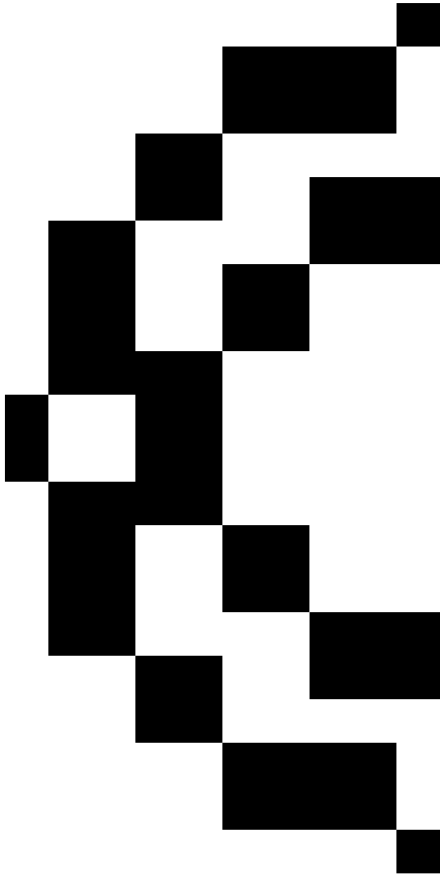
```
msg = "Universidade NOVA de Lisboa"
upper_msg = msg.upper()
lower_msg = msg.lower()
print(msg)        # Universidade NOVA de Lisboa
print(upper_msg)  # UNIVERSIDADE NOVA DE LISBOA
print(lower_msg)  # universidade nova de lisboa
```

Exercise 3

Define `my_string = "patient"`. Without hardcoding any index number, calculate the position of the middle character using `len()` and `int()`. Store the result in a variable `middle_index` and print the character at that position.

Example

- `Input:  my_string = "patient"`
- `Output: i`



Exercise 3 — Solution

```
my_string = "patient"
middle_index = int(len(my_string) / 2)  # int(7 / 2) = int(3.5) = 3
print(my_string[middle_index])          # "i"
```

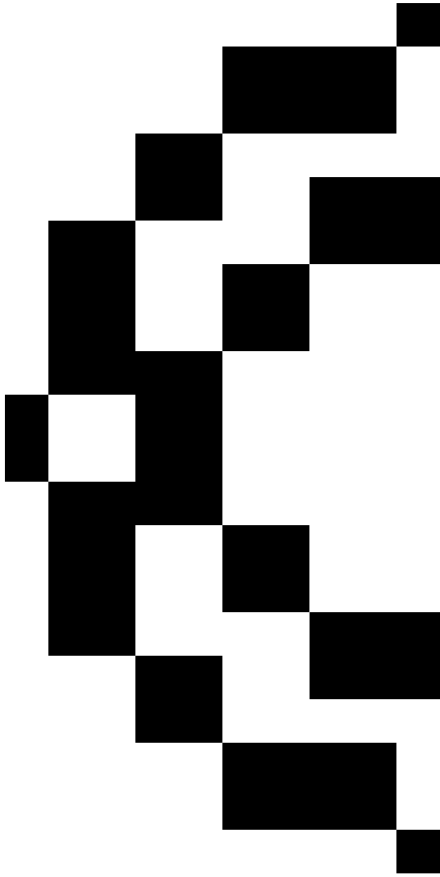
Exercise 4

Three variables are defined below. Using only these variables — no other numeric literals — write four print statements that produce the outputs listed.

```
one = "18"
two = "44"
three = "22"
```

Required outputs (one per line):

```
1844
2222
66
20
```



Example

```
Input:  one = "18", two = "44", three = "22"
Output: 1844
        2222
        66
        20
```

Exercise 4 — Solution

```
one = "18"
two = "44"
three = "22"

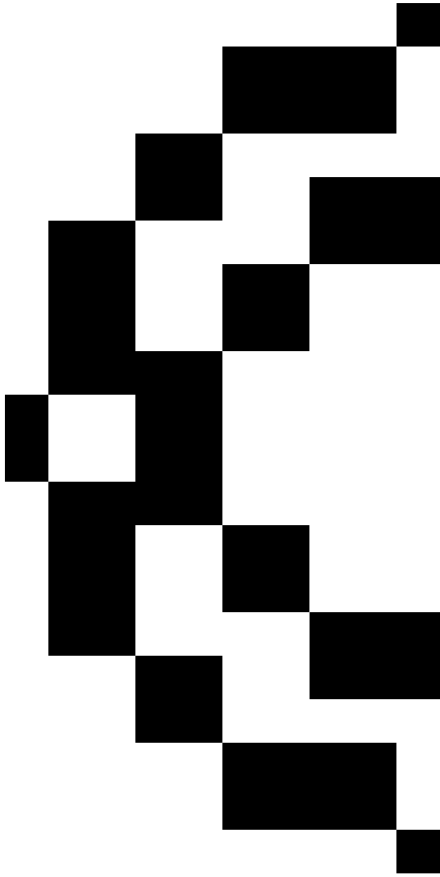
print(one + two)                  # "1844" — string concatenation
print(three + three)              # "2222" — string concatenation
print(int(two) + int(three))      # 66    — numeric addition (44 + 22)
print(int(one) + int(three[0]))   # 20    — int("18") + int("2") = 18 + 2
```

Exercise 5

Create a list `my_list` containing the strings `"A"`, `"B"`, `"C"`, `"D"`, `"E"`.
Print the element at index 2. Change the value at index 2 to `"X"` and print it again.
Then create a second list `list_two` containing `"F"`, `"G"`, `"H"`. Combine both lists into a new list `new_list` and print its length.

Example

- `Input:  (as described above)`
- `Output: C`
- `X`
- `8`



Exercise 5 — Solution

```
my_list = ["A", "B", "C", "D", "E"]
print(my_list[2])    # C
my_list[2] = "X"
print(my_list[2])    # X

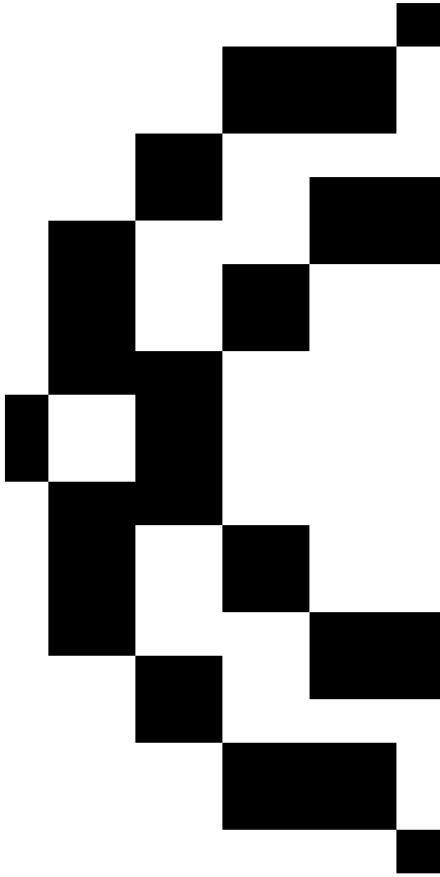
list_two = ["F", "G", "H"]
new_list = my_list + list_two   # ["A", "B", "X", "D", "E", "F", "G", "H"]
print(len(new_list))            # 8
```

Exercise 6

Define `my_string = "Gertrude bought 3 sodas for 1.20€ each."`. Without modifying the string, extract the number of sodas and the price per soda using string indexes. Convert them to the appropriate numeric types, multiply them to calculate the total, and print it.
You will notice the result is not exactly `3.6`. Print a second line with the total rounded to two decimal places using `round(total, 2)`.

Example

```
Input:  my_string = "Gertrude bought 3 sodas for 1.20€ each."
Output: Gertrude paid a total of 3.5999999999999996
        Gertrude paid a total of 3.6
```



Exercise 6 — Solution

```
my_string = "Gertrude bought 3 sodas for 1.20€ each."
num_sodas = int(my_string[16])        # "3" → 3
price = float(my_string[28:32])       # "1.20" → 1.2
total = num_sodas * price
print("Gertrude paid a total of", total)           # 3.5999999999999996
print("Gertrude paid a total of", round(total, 2)) # 3.6
```

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Thank you

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